

HOW We Are Doing This Research

Exact execution workflow - from raw solar data to final scientific result

Research Project: Cross-Cycle Explanation Drift in Solar Flare Forecasting

Domain: Space Weather | Machine Learning | Explainable AI

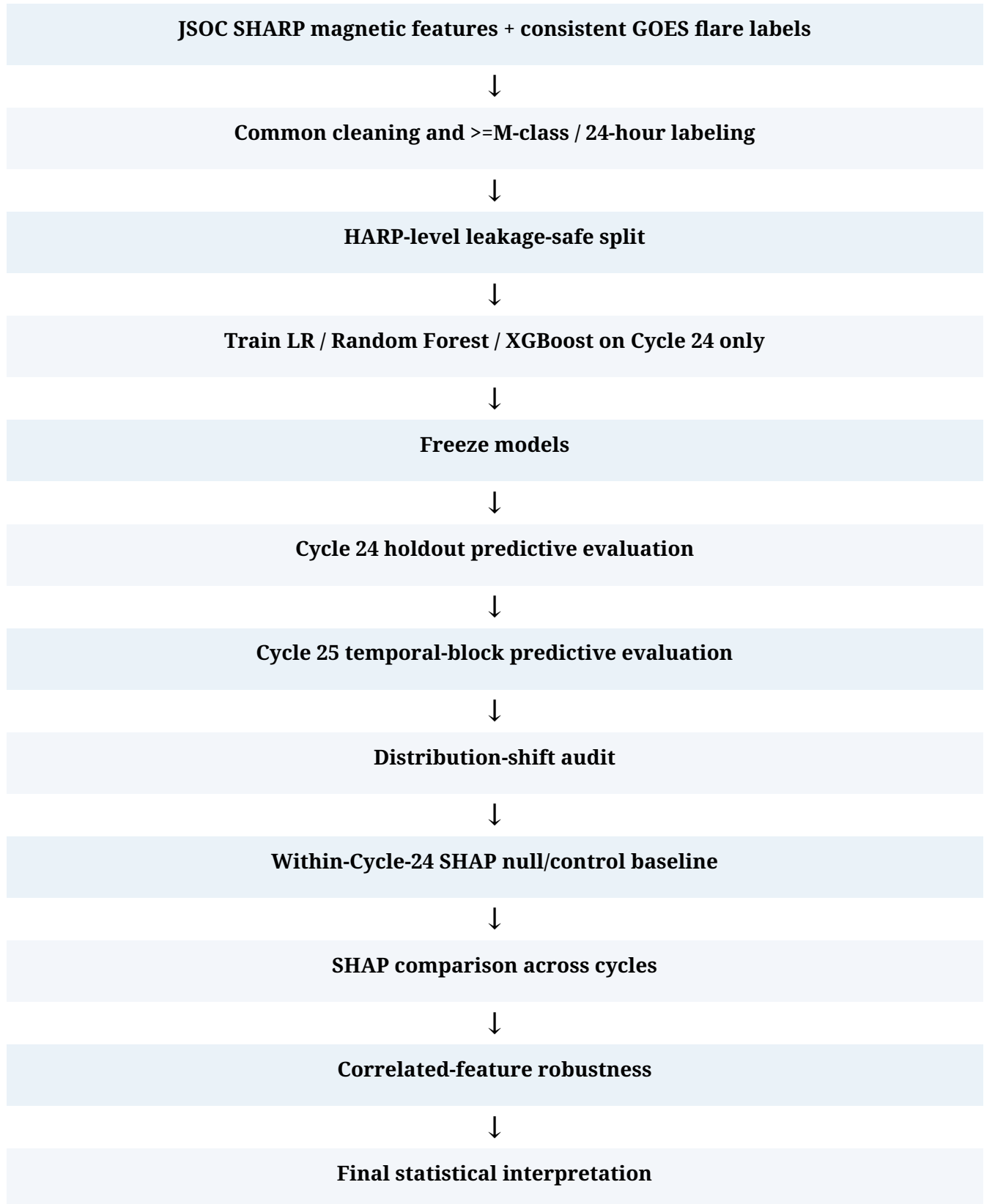
Status: Locked research direction - October 2026

Purpose

Use this document as the practical roadmap. It defines the seven execution phases, their purpose, and the controls required before interpreting explanation drift.

Based on the locked Master Research Reference v1.0

1. Full Pipeline at a Glance



Phase 1 - Data Collection and Dataset Design

WHAT: Cycle 24 aur Cycle 25 ke liye SHARP magnetic parameters aur GOES flare labels collect karke one consistent dataset design banayenge.

WHY: Fair cross-cycle comparison tabhi possible hai jab input definitions, labels aur provenance dono cycles mein consistent ho.

HOW: JSOC se SHARP features query karenge. One documented GOES catalog source se flare events lenge. Target: next defined forecast window (initially 24 hours) mein $\geq M$ -class flare = 1, otherwise 0. Exact query/version information save karenge.

Phase output

Versioned raw-data queries + feature table + flare-event table + documented labeling rule.

Phase 2 - Cleaning, Preprocessing and Leakage-Safe Splitting

WHAT: Quality filtering, missing-value handling, limb cut, scaling/imputation, aur active-region grouping implement karenge.

WHY: Same AR train aur test dono mein aa gaya toh performance artificially inflate ho sakti hai. Preprocessing leakage bhi result ko invalid bana sakta hai.

HOW: Split HARP/active-region level par. No AR overlap across train/validation/test. Imputation, scaling and class-weighting rules Cycle-24 training set se fit honge; Cycle 24 holdout aur Cycle 25 par only transform/apply hoga. Limb cut (e.g. ± 70 deg) sensitivity test ke saath justify karenge.

Phase output

Leakage-safe split manifest + fitted preprocessing pipeline + clean Cycle-24/Cycle-25 analysis tables.

Phase 3 - Cycle 24 Model Training and Baseline

WHAT: Logistic Regression, Random Forest aur XGBoost ko sirf Cycle-24 training ARs par train karenge, then Cycle-24 validation/holdout se baseline establish karenge.

WHY: Cycle 25 ko truly unseen transfer regime rakhna hai. Baseline ke bina cross-cycle performance change quantify nahi kar sakte.

HOW: Hyperparameters Cycle-24 validation data se select. Final models freeze. Independent Cycle-24 holdout par TSS primary metric; HSS/F1, PR-AUC and calibration/Brier score as complementary metrics. Confidence intervals active-region-level bootstrap se.

Phase output

Three frozen models + Cycle-24 predictive baseline with uncertainty.

Phase 4 - Frozen-Model Evaluation on Solar Cycle 25

WHAT: Same trained models ko bina retraining/fine-tuning Cycle 25 temporal blocks par evaluate karenge.

WHY: Yahi cross-cycle predictive generalization experiment hai.

HOW: Same preprocessing transform, same feature definitions and same threshold-selection rule use karenge. Cycle 25 ko temporal blocks mein evaluate karke performance and calibration change track karenge.

Phase output

Cycle-25 predictive results + predictive-drift comparison against Cycle-24 holdout.

Phase 5 - Distribution-Shift Audit and Null Baseline

WHAT: Cycle 24 vs 25 feature distributions/label prevalence compare karenge aur Cycle 24 ke andar normal SHAP variability estimate karenge.

WHY: Observed SHAP change ko meaningful bolne se pehle normal sample fluctuation aur obvious data shift se compare karna necessary hai.

HOW: Feature distribution plots/tests, missingness, AR properties and positive-label prevalence audit. Cycle-24 holdout ko repeated AR-level bootstrap/resampling se compare karke within-cycle SHAP variability band build karenge.

Phase output

Distribution-shift report + within-Cycle-24 null/control distribution for explanation variability.

Phase 6 - SHAP Explanation-Drift Analysis

WHAT: Same frozen model ki Cycle-24 holdout aur Cycle-25 SHAP attributions compare karenge.

WHY: Research ki main novelty predictive score aur explanation structure ko separate dimensions ke roop mein test karna hai.

HOW: Fixed documented SHAP background/reference from Cycle-24 training data. Global/local attributions calculate. Metrics: mean |SHAP| change, sign/direction consistency, Spearman, Kendall, top-k overlap, attribution-distribution distances. Multiple-comparison correction apply. Strongly correlated SHAP features ko clusters/groups mein bhi analyze karenge.

Phase output

Per-model explanation-stability statistics + significance relative to within-cycle baseline + correlated-feature robustness results.

Phase 7 - Final Scientific Interpretation and Reproducibility

WHAT: Predictive and explanation results combine karke evidence-based conclusion banayenge aur full reproducible experiment package preserve karenge.

WHY: Result ko overclaiming se bachana aur independent reproduction possible banana professional research ke liye essential hai.

HOW: Four outcome matrix use karenge: stable/stable, stable/drift, drop/drift, drop/stable. SHAP ko causal solar physics proof nahi bolenge. Code, split manifests, data-query scripts, random seeds, environment file and figure-generation scripts package karenge.

Phase output

Defensible scientific conclusion + reproducible research package ready for manuscript stage.

2. Critical Rules During Execution

- Cycle 25 par model retrain/fine-tune nahi karna in the primary experiment.
- Same active region train and test groups mein kabhi nahi aana chahiye.
- Preprocessing Cycle-24 training data par fit hona chahiye.
- Same threshold-selection rule cross-cycle evaluations mein preserve karna.
- SHAP comparison same fixed model across cycles ka hona chahiye, model A vs model B ka primary evidence nahi.
- Cross-cycle explanation drift ko within-cycle null baseline se compare kiye bina declare nahi karna.
- Correlated feature rank swaps ko cautiously interpret karna.
- SHAP = model attribution, not causal solar physics.

3. Final Decision Logic

Evidence pattern	Research interpretation
TSS similar + SHAP within null band	No strong evidence of explanation drift.
TSS similar + SHAP beyond null band	Evidence for explanation drift despite similar predictive skill; inspect covariate/prevalence causes.
TSS drops + SHAP beyond null band	Both predictive and attribution structures are unstable under cross-cycle shift.
TSS drops + SHAP within null band	Prediction degrades while attribution structure remains comparatively stable; investigate calibration/prevalence/range shift.

Shortest HOW

Collect -> clean -> HARP-safe split -> train on Cycle 24 -> freeze -> test Cycle 24 -> test Cycle 25 -> audit data shift -> build Cycle-24 SHAP null baseline -> compare SHAP -> test correlated-feature robustness -> interpret.